

MM-ACT: Learn from Multimodal Parallel Generation to Act

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Abstract

A generalist robotic policy needs both semantic understanding for task planning and the ability to interact with the environment through predictive capabilities. To tackle this, we present MM-ACT, a unified Vision-Language-Action (VLA) model that integrates text, image, and action in shared token space and performs generation across all three modalities. MM-ACT adopts a re-mask parallel decoding strategy for text and image generation, and employs a one-step parallel decoding strategy for action generation to improve efficiency. We introduce Context-Shared Multimodal Learning, a unified training paradigm that supervises generation in all three modalities from a shared context, enhancing action generation through cross-modal learning. Experiments were conducted on the LIBERO simulation and Franka real-robot setups as well as RoboTwin2.0 to assess in-domain and out-of-domain performances respectively. Our approach achieves a success rate of 96.3% on LIBERO, 72.0% across three tasks of real Franka, and 52.38% across eight bimanual tasks of RoboTwin2.0 with an additional gain of 9.25% from cross-modal learning. We release our codes, models and data at <https://github.com/HHYHRHY/MM-ACT>.

1. Introduction

A generalist robotic policy requires both high-level semantic understanding and the ability to interact effectively with the environment. In recent years, Vision-Language-Action (VLA) models [3, 19–21, 24, 28, 46] have emerged as a promising paradigm for building generalist robotic policies. These approaches typically build upon large-scale pretrained Vision-Language Models (VLMs) by integrating action heads or expert modules to bridge perception and

control. However, while the underlying VLMs excel at visual and semantic understanding, they often lack an explicit model of physical dynamics [15, 45, 51], which limits their ability to guide temporal action generation [10, 54].

Other works [7, 14, 18, 35, 59] further extend the conventional state-to-action imitation learning paradigm into a visual-prediction-driven decision and planning framework. By incorporating visual prediction into the policy learning process, these approaches enable models to explicitly or implicitly model future visual dynamics, thereby achieving stronger predictability and planning capabilities in complex interactive environments. Although these world models [4, 37] excel in temporal and environmental dynamics, they are primarily trained for predictive objectives rather than task-oriented planning. This leads to limited instruction understanding and sub-task planning capabilities.

Recent unified VLA approaches have largely inherited the development paradigm of unified understanding and generation models [11, 34, 39, 43]. Instead of rethinking the policy architecture, these methods design action generation by closely following the base model’s modeling paradigm. For example, some works like [50] retain the autoregressive text generation paradigm while adopting parallel decoding strategy for image and action generation, forcing the model to learn to handle single-token prediction and block-level tokens prediction in forward process. This in turn requires multiple attention mechanisms and substantially increases both architectural and training pipeline complexity. Others such as [6, 40] adopt a fully autoregressive generation paradigm for text, image and action generation, which leads to slow inference speed for action generation.

To address these challenges, we propose MM-ACT, a unified model that jointly generates text, image, and action using a parallel decoding strategy. MM-ACT integrates text, image and action into a unified sequence of discrete tokens through modality-specific tokenizers, and learns to predict task planning, future image prediction and action chunk using block-level masked token prediction. For text

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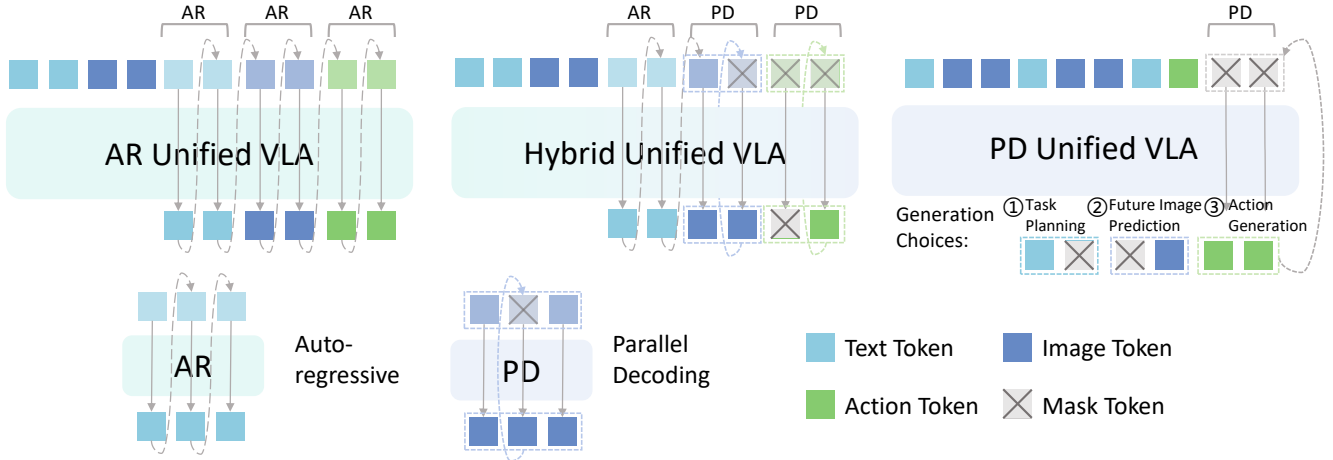


Figure 1. **Comparison of different unified VLA paradigms.** (a) **Autoregressive (AR) unified VLA [50]:** All three modalities—text, image, and action—are generated using an autoregressive approach. (b) **Hybrid unified VLA [6, 40]:** Text is generated autoregressively, while image and action are generated using a re-mask parallel decoding approach. (c) **Parallel decoding (PD) unified VLA (Ours):** Converts images and instructions into a multimodal interleaved input, with text, image, and action generated through parallel decoding. This approach supports three separate generation settings: ① sub-task planning, ② future image prediction, and ③ action generation. ① ② adopt a multiple-step parallel decoding strategy with re-masking, while ③ utilizes a one-step parallel decoding strategy.

and image generation, MM-ACT adopts a re-mask parallel decoding strategy, while for action generation it employs a one-step parallel decoding strategy to achieve low-latency inference. We further analyze the trade-offs between these two strategies in terms of effectiveness and efficiency. Figure 1 compares our parallel-decoding unified VLA with prior autoregressive and hybrid unified VLA paradigms. In contrast to prior approaches that rely on autoregressive text modeling, MM-ACT uses bidirectional attention over the full multimodal sequence, providing a more unified architecture that simplifies the training pipeline. Rather than combining modality-specific decoders, MM-ACT fully unifies text, image, and action by representing them as equivalent discrete tokens and optimizes them with a masked-token objective under same decoding principle without modality specific attention or decoder design.

Building upon this unified modeling, we further propose **Context-Shared Multimodal Learning**, a training pipeline for three modalities generation. For identical multimodal context from robot’s current views, task instructions, text descriptions and robot’s states, we annotate data across the following categories: task planning, which involves planning the overall task and determining which sub-task to execute; future image prediction, consisting of image that represent the completion image of executed action chunk; and action chunk, referring to the chunk-step actions from expert data at the current timestep t . After initial training steps on text and image generation, we further perform forward processes for all three modalities under the shared context, aggregating their losses within same gradient accumulation step for optimization. During deployment, the

model performs action-only generation with one-step decoding for efficiency. Since inference does not require additional multimodal generation, our method reduces latency and runs stably at a 40Hz action rate.

Our experiments aim to validate the feasibility of our modeling approach and the action generation’s enhancement via our training pipeline. In simulation experiments, MM-ACT achieves an average of 96.3% performance on Libero, 52.38% in 8 tasks on RoboTwin2.0 and 72.0% on Franka real-world experiments, consistently outperforming overall baselines. Moreover, our proposed training pipeline achieves a 9.25% improvement in action generation performance compared to action-only learning baseline. Additionally, we demonstrate that either text-action or image-action joint training can also yield performance gains.

2. Related Work

2.1. Discrete Diffusion Language Model

Discrete diffusion models have recently emerged as promising alternatives to autoregressive language models. Unlike token-by-token decoding in AR models, diffusion-based approaches generate sequences through iterative denoising of corrupted inputs. Early models [1, 17] demonstrated the feasibility of applying discrete denoising to text. Subsequent work [30] introduced improvements in masking strategies and training objectives. Large-scale bidirectional diffusion transformers [27, 44] further advanced performance, achieving competitive results across text and multimodal tasks. These efforts establish discrete diffusion as a viable and increasingly competitive generation paradigm.

In this work, we explore extending discrete diffusion models to the domain of action parallel decoding.

2.2. Unified Vision-Language Models

Autoregressive architectures have long been dominant in multimodal understanding, while diffusion-based models have become fundamental for image generation. Recently, there has been an increasing focus on developing unified frameworks [53] that combine these tasks. Unified models’ architectures can generally be abstracted into three components: modality-specific encoder, modality-fusion backbone, and modality-specific decoder. For autoregressive architectures [13, 16, 34, 39, 42], both visual and language tokens are processed in a sequential manner, the modality-fusion backbone autoregressively predicts multimodal outputs. For diffusion-based architectures [22, 32, 38, 44], the denoising process is extended from timestep and noise to incorporate multimodal contexts, such as textual content, images, or joint embeddings. In hybrid autoregressive and diffusion architectures [11, 43, 57], text tokens are generated autoregressively, while image tokens are generated through a multi-step denoising process. We aim to unify the training objectives and attention mechanisms for multimodal generation to simplify the training design. At the same time, we aim to maintain fast action generation by adopting parallel decoding for actions. Therefore, we have chosen MMaDA [44] as the base model.

2.3. Vision-Language-Action Models

The Vision-Language-Action (VLA) model primarily aims to transform general visual-language inputs into a sequence of executable action outputs within a unified framework. Early methods built upon large-scale pre-trained Vision-Language Models (VLMs) by incorporating action heads or specialized action expert modules [3, 5, 19–21, 28, 33, 60]. This allowed the model to simultaneously possess visual perception, language understanding, and action decision-making capabilities—achieving an end-to-end integration of vision, language, and action in one unified system. To bridge the latency gap between VLMs and the real-time demands of action systems, a dual-system design [2, 24, 29, 31, 49] is commonly adopted, decoupling the overall architecture into an upstream VLM and a downstream action-prediction head. The latter either directly decodes actions from the latent action tokens generated by the VLM, or employs an additional diffusion-based action head to the VLM’s outputs or intermediate hidden representations. Various reasoning strategies [9, 12, 46, 48, 58] have also been used to augment the action model, ranging from purely language-based reasoning to multimodal reasoning capabilities. However, these approaches typically perform diffusion-based fine-tuning for action generation directly on top of autoregressive (AR) pre-trained backbones, result-

ing in an objective misalignment between AR-based pre-training (token prediction) and diffusion-based fine-tuning (denoising). This inconsistency can introduce optimization misalignment and hinder the model’s ability to effectively leverage its pre-trained knowledge. Based on [27], [23] introduces discrete diffusion into action modeling and demonstrating the effectiveness of this approach.

In contrast to the above methods, our proposed model is built upon a diffusion unified model [44] (dLLM) as the foundation for the VLA. This design ensures that the model follows a consistent parallel decoding based generation objective during all training stages, thereby achieving better control and generalization capabilities.

3. Method

3.1. Model Design

We leverage an 8B-parameter Transformer-based [36] mask tokens predictor equipped with bidirectional attention mechanisms to facilitate generation tasks across three modalities. Our model represents text, image and robot’s proprioceptive states as a single sequence of discrete tokens, drawn from the concatenated vocabularies of three modality-specific tokenizers. We add a modal token before the context that specifies the target generation modality, append a fixed-length `<mask>` token block after the context. At inference time, we adopt different decoding strategies and compute probabilities at specific tokenizer positions corresponding to the modal token. Following MMaDA [44], we use the LLaDA [27] model’s tokenizer for text and employ pretrained image quantizer from Showo [43], encoding and decoding image inputs and outputs, which uses 8,192 tokens as the image codebook. For image inputs, we first pad each image to a square, downsample it to 256×256 and encode it into 256 tokens. For image generation, the model outputs 256 tokens, and the image quantizer decodes them into a 256×256 image. For robot’s states inputs and action outputs, we adopt the bin tokenizer [19] as the quantization method, allocating 2,048 tokens dedicated to action generation. On the input side, each continuous scalar is first normed to the range of [-1, 1], then is quantized to a token in the codebook; on the output side, tokens are detokenized back into continuous scalars to represent continuous action values. We concatenate action codebook to the end of the tokenizers, without affecting the original text tokenizer and image codebook.

3.2. Context-Shared Multimodal Learning

Context-Shared Multimodal Input We use a shared context for the generation tasks across the three modalities. The context for each modality is defined as $C_{\text{modal}} = \text{<modal>} + \text{sharedinput}$, where the modal token $\text{<modal>} \in \{\text{<|mm2a|>}, \text{<|mmu|>}, \text{<|t2i|>}\}$.

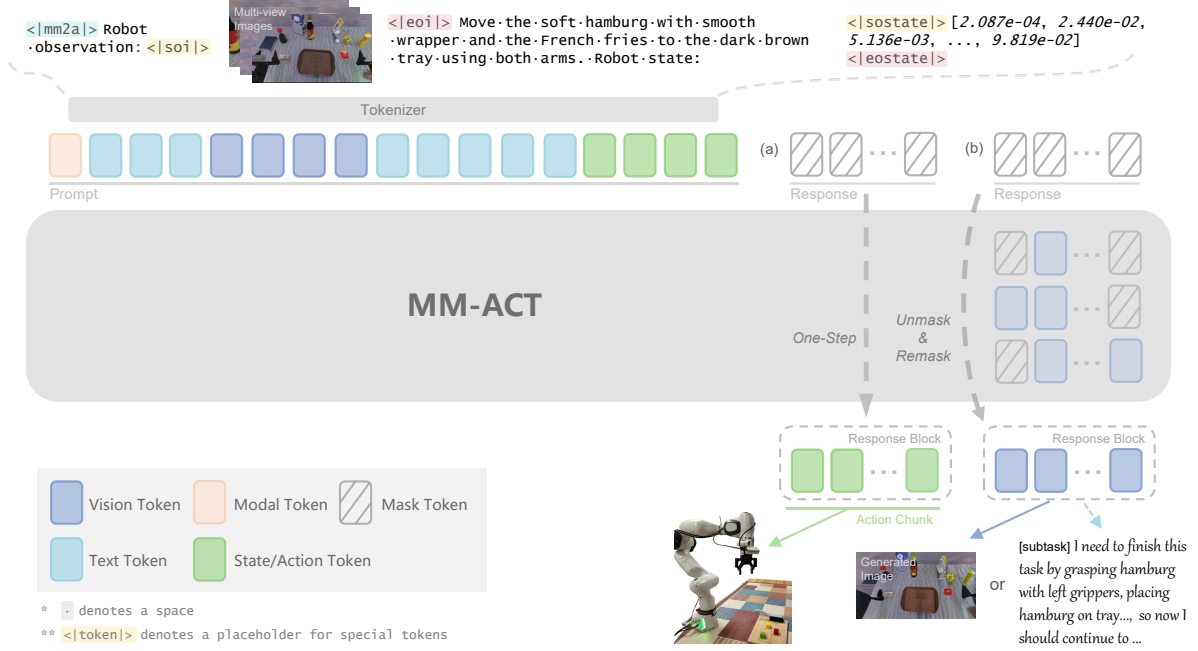


Figure 2. **The architecture of MM-ACT.** MM-ACT uses modality-specific tokenizers to tokenize text, image, and action into discrete tokens within a shared space. Given the shared multimodal input, the model determines whether to execute task planning, future image prediction, or action generation based on the modal token, with each task corresponding to the generation of text, image, or action.

sharedinput is a modality-interleaved token sequence that follows a predefined template to convert the inputs of robot’s multi-view observations, task instructions, text descriptions, and optionally the robot’s states.

We append a fixed-length block for image and action modal after C_{modal} . Specifically, the text block size is set to 256 to accommodate task planning sequence. We concatenate variable-length textual annotations after the context and append $\langle \text{eos} \rangle$ tokens to match the maximum sequence length. The image block size is also 256, enabling the generation of a single image. The action block size $N_{\text{act_block}} = d_{\text{action}} * N_{\text{chunk_size}}$, where d_{action} is the dimension of action and $N_{\text{chunk_size}}$ is the number of actions in one chunk. It enables chunk-wise action generation, with $N_{\text{chunk_size}}$ kept fixed during both training and inference.

We set the same maximum sequence length for the inputs of all three modalities, and shorter sequences are padded with the $\langle \text{pad} \rangle$ token to match this maximum length.

Multimodal Learning with Unified Objective The generation tasks across the three modalities share a common context, and we append modality-specific blocks with mask tokens after the context to train generation capability. Unlike approaches that combine multiple objectives, such as autoregressive text generation and diffusion-based generation for images [57] or actions [41], we adopt the same optimization objective to train generation across all three modalities. We model the specific block of each modality

as a token sequence $x_0 = (x_0^1, x_0^2, \dots, x_0^L)$, L is the sequence length. For a continuous time $t \in (0, 1]$, construct a masked sequence x_t by independently masking each position with probability $p_{\text{mask}} = f_{\text{modal}}(t)$, where f_{modal} is the function of mask schedule in each modal. The conditional distribution of x_t^i can be formed as:

$$q_t(x_t^i | f_{\text{modal}}(t), x_0^i) = (1 - f_{\text{modal}}(t)) \mathbf{1}\{x_t^i = x_0^i\} + f_{\text{modal}}(t) \mathbf{1}\{x_t^i = \langle \text{mask} \rangle\}. \quad (1)$$

We use a linear schedule for text modality followed by LLaDA [27], and a cosine schedule for both the image and action modalities to better align with the noise schedule for continuous denoising. The conditional distribution of x_t can then be written as:

$$q_t(x_t | x_0) = \prod_{i=1}^L q_t(x_t^i | f_{\text{modal}}(t), x_0^i) \quad (2)$$

We set $t = 1$ for action modal to train the model generating all tokens from a fully masked sequence in a single forward pass, the masked sequence $x_t = \langle \text{mask} \rangle \times L$.

Formally, MM-ACT is trained as a mask token predictor, a parametric model $p_{\theta}(\cdot | C_{\text{modal}}, x_t)$ that takes C_{modal} and x_t as inputs and predicts all masked tokens simultaneously. We define a unified cross-entropy loss over the masked tokens across the three modalities.

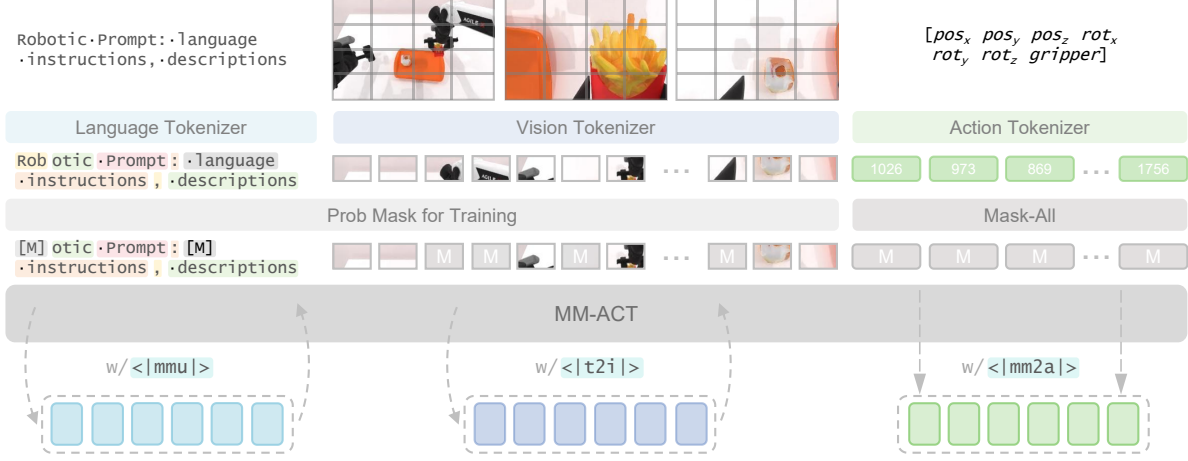


Figure 3. **Training pipeline of MM-ACT.** Within the shared context, ground truth of three modalities are masked according to the decoding strategy of each modality, and appended to the context. The model takes these inputs to perform forward processes across three modality generation tasks, computing the loss specifically on the masked tokens.

$$\mathcal{L}(\theta) = -\mathbb{E}_{t, x_0, x_t} \left[\sum_{\text{modal} \in \mathcal{M}} \frac{\lambda_{\text{modal}}}{t} \sum_{i \in \mathcal{I}_{\text{modal}}} \mathbf{1}\{x_t^i = \text{M}\} \times \log p_{\theta}(x_0^i \mid C_{\text{modal}}, x_t) \right]. \quad (3)$$

where x_0 is sampled from the training data, x_t is sampled from $q_t(x_t \mid x_0)$, M is `<mask>` token. The indicator function $\mathbf{1}[\cdot]$ ensures that the cross-entropy loss is computed only for masked tokens. $\mathcal{M} = \{\text{<|mm2a|>, <|mmu|>, <|t2i|>}\}$, and λ_{modal} represents the weight of each modal’s loss, used to control the impact of each modal on model optimization during training process. In Figure 3, we provide an illustration of our mask data construction process during training.

We adopt a two-stage training strategy. In Stage 1, we set λ_{mm2a} to 0 and initially train the model exclusively on text and image generation tasks, optimizing until the losses for text and image modalities reach low values. In Stage 2, we primarily focus on supervising action generation, adjusting λ_{mmu} and λ_{t2i} to approximately 0.05–0.1 to maintain their generative capabilities.

Parallel Decoding Strategy We formulate the model as a block-level masked-token predictor. For action generation, we produce all action tokens in a single forward pass to maintain efficiency. We also introduce re-mask decoding strategy for action generation in our design, adopt a low-confidence remasking strategy and use a cosine noise schedule consistent with MAGVIT-v2 [47]. We compare effectiveness and efficiency in Appendix A.2. We employ re-mask decoding strategy for image generation, with same re-mask strategy and noise schedule to action.

For text generation, we limit the generation sequence length to 256 tokens. This constraint is adopted because task planning annotations in our manipulation tasks can typically be completed within this range. Furthermore, this limit aligns with the default block size of LLaDA [27], allowing us to restrict the entire generation process to a single block. As a result, we do not employ the semi-autoregressive approach for text generation. Instead, text, image, and action generation are all performed within one block. Logits are computed for all masked positions, and a subset of tokens selected either randomly or based on confidence scores is predicted. The masking schedule is linear, and the forward process is repeated for a fixed number of steps. More details can be found in Appendix C.

4. Experiments

Our experiments center around two primary questions: (1) whether our model architecture can effectively perform action generation to accomplish manipulation tasks in both in-domain and out-of-domain settings; and (2) what kind of improvements our training pipeline provides across generation tasks in different modalities.

In 4.1, we describe the benchmarks and experimental setups used in both simulation and real-robot environments. 4.2 introduces the datasets and training details. 4.3 presents the evaluation results for action generation. 4.4 provides an analysis of how our training pipeline improves action generation and how the image modality is affected under the training pipeline. Additional details about experiments of modalities effects and ablation studies on decoding strategies of each modalities and robot’s state in text or image’s context can be found in Appendix A.

Model	Spatial SR (%)	Object SR (%)	Goal SR (%)	Long SR (%)	Average SR (%)
OpenVLA [19]	84.7	88.4	79.2	53.7	76.5
π_0 + FAST [28]	96.4	96.8	88.6	60.2	85.5
π_0 [3]	96.8	<u>98.8</u>	<u>95.8</u>	85.2	94.2
OpenVLA-OFT [20]	96.2	98.3	96.2	90.7	95.4
CoT-VLA [55]	87.5	91.6	87.6	69.0	81.1
TraceVLA[56]	84.6	85.2	75.1	54.1	74.8
DreamVLA [52]	<u>97.5</u>	94.0	89.5	89.5	92.6
WorldVLA (512*512) [6]	87.6	96.2	83.4	60.0	81.8
UniVLA [40]	95.4	<u>98.8</u>	93.6	94.0	<u>95.5</u>
MM-ACT (Vanilla)	97.8	99.4	94.8	88.0	95.0
MM-ACT (+Text in Long)	-	-	-	<u>93.0</u> (+5.0%)	96.3

Table 1. **LIBERO task performance results (%)**. **Bold** values denote the best performance, and underlined values denote the second-best.

Model	Adjust Bottle	Beat Hammer Block	Click Bell	Dump Bin Bigbin
π_0 [3]	89%	<u>68%</u>	40%	61%
OpenVLA-OFT [20]	64%	7%	24%	<u>31%</u>
MM-ACT (Vanilla)	51%	61%	86%	13%
MM-ACT (+Text)	<u>75%</u>	67%	<u>91%</u>	13%
MM-ACT (+Image)	72%	64%	<u>91%</u>	8%
MM-ACT (+Text&Image)	71%	78%	95%	13%
Model	Move Playingcard Away	Lift Pot	Place Burger Fries	Place Can Basket
π_0 [3]	47%	29%	41%	10%
OpenVLA-OFT [20]	17%	3%	31%	8%
MM-ACT (Vanilla)	30%	40%	46%	<u>18%</u>
MM-ACT (+Text)	24%	28%	56%	<u>18%</u>
MM-ACT (+Image)	<u>39%</u>	<u>31%</u>	<u>72%</u>	13%
MM-ACT (+Text&Image)	39%	<u>31%</u>	73%	19%
Overall Avg: π_0 = 48.13%, OpenVLA-OFT = 23.13%, MM-ACT (Vanilla) = 43.13%				
MM-ACT (+Text) = 46.5%(+3.37%) MM-ACT (+Image) = <u>48.75%</u> (+5.62%) MM-ACT (+Text&Image) = 52.38% (+9.25%)				

Table 2. **RoboTwin task performance results**. **Bold** values denote the best performance, and underlined values denote the second-best.

4.1. Benchmark

Simulation Benchmark We conducted validation of diverse tasks in the simulation experiments.

LIBERO [25] is a benchmark based on the Franka robotic arm, designed to evaluate lifelong learning and knowledge transfer capabilities in robots. In this work, we adopt four sub-benchmarks from LIBERO: Libero-Spatial for spatial reasoning, Libero-Object for object-centric understanding, Libero-Goal for goal-conditioned variations, and Libero-Long for long-horizon, compositional tasks. Each sub-benchmark contains 10 distinct tasks, and each task is provided with 50 tele-operated demonstrations.

RoboTwin [8, 26] is a simulation benchmark designed for bimanual robotic manipulation. We employ Robotwin2.0 [8] for our experiments. This simulation platform provides a multi-task data collection and evaluation framework across multiple robot embodiments. By introducing a large variety of scenes and objects for domain randomization, it both scales up the available robot training

data and enables out-of-domain, unseen evaluation. In this work, we adopt the Agilex Piper dual-arm setup and evaluate on eight representative tasks in unseen settings, where the task instructions, environments and object locations are all unseen in training datasets.

Real-world Experiments We employ Franka real-world experiments for testing the in-domain action generation ability among our model and baselines. We use an Intel RealSense D435 camera mounted on the wrist and an Intel RealSense D435i camera positioned as a third-person view to provide visual observations for the model. We design three manipulation tasks—press button, stack small block on big block, and sort vegetables and fruits (see Figure 4 for task illustrations). For each task and each model, we conduct 20 evaluation trials. For different tasks, we employ distinct success rate calculation methods. For press button and stack block, the task is considered successful when the entire task is completed. For sort vegetables and fruits, the success rate is calculated based on the number of objects

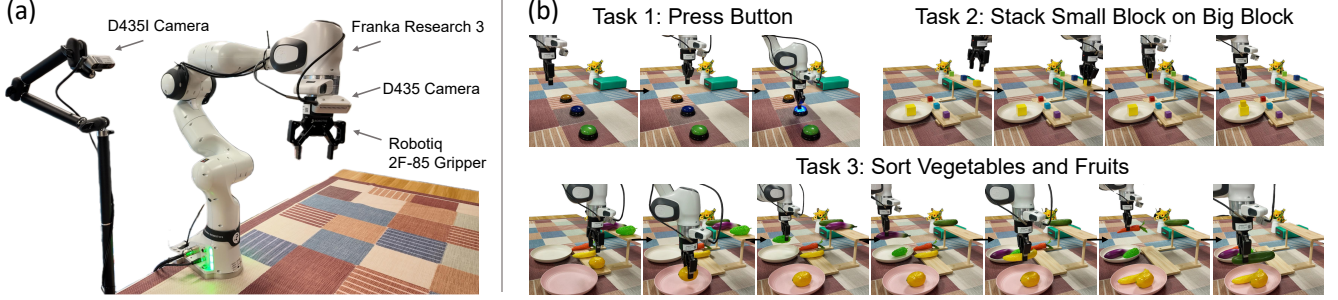


Figure 4. (a) **Franka real-world setup.** The Franka Research 3 robot, equipped with a D435i camera as an external camera, a D435 camera as a wrist camera, and a Robotiq 2F-85 gripper, is shown in the experimental setup. (b) **Task examples.** **Press Button:** The robot presses the blue button. **Stack Small Block on Big Block:** The robot stacks the small yellow block on top of the large yellow block. **Sort Vegetables and Fruits:** The robot places fruits from the rack into the right plate and vegetables into the left plate.

correctly placed in the right spot.

Baselines We evaluate our method against three major paradigms: (1) VLM-based VLA (e.g. OpenVLA [19], OpenVLA-OFT [20], π_0 [3]), which focus primarily on semantic understanding but generally lack dynamic modeling; (2) Visual Prediction VLA (e.g. CoT-VLA [55], TraceVLA [56], DreamVLA [52]), which emphasize future prediction but have relatively limited task reasoning; and (3) Unified VLA based on the unified model architecture (e.g. UniVLA [40], WorldVLA [6]), which possess the ability to generate multiple modalities and is capable of performing visual or textual reasoning.

4.2. Implementation Details

Datasets For LIBERO benchmark, we use the official datasets provided from the benchmark. To verify whether our training pipeline can leverage text-based task planning to improve action generation, we focus on the long-horizon LIBERO-Long setting. We manually decompose each of the 10 tasks into a sequence of subtasks, annotate keyframes corresponding to subtask transitions for every episode, and then generate subtask planning labels for each frame using a template-based scheme.

For RoboTwin2.0 benchmark, we collect 500 expert episodes per task under the domain-randomized setting and automatically filter out trajectories exhibiting undesirable behaviors such as overly long pauses, resulting in a total of about 70k training samples. We further design an automatic subtask-annotation framework on simulation platform, based on the rule-based trajectory generation used for collecting RoboTwin expert data. We timestamp each invocation of motion, grasp, and place primitives from the skill library, and attach task-specific language descriptions for each skill function called in each task. This allows us to automatically obtain the executed action chunk, the corresponding subtask language annotation and the future image after action execution for every frame.

For each Franka real-robot task, we collect 100 demonstration trajectories via manual teleoperation.

All of our model weights are directly trained from the base weights of MMaDA [44] on the above datasets. More details of our datasets can be found in Appendix B.

Training Details We train our model with batch size of 128 and action chunk size of 8 across three experiments. For LIBERO benchmarks, we train four separate models, one per benchmark, for about 15k steps each. For eight RoboTwin tasks, we adopt multi-task training and optimize action generation for about 27k steps. In the Franka real-robot experiments, we train for roughly 8k steps per task. More details can be found at Appendix A.3.

4.3. Evaluation Results

Benchmark Results As shown in Table 1, MM-ACT achieves the best average success rate of 96.3%, surpassing all existing baselines. Specifically, the vanilla version of MM-ACT achieves success rates of 97.8%, 99.4%, 94.8%, and 88.0% on Libero-Spatial, Libero-Object, Libero-Goal, and Libero-Long respectively. For VLM-based VLA, our approach outperforms OpenVLA by 19.8%, π_0 + FAST by 10.8%, π_0 by 2.1%, and OpenVLA-OFT by 0.9%. For Visual Prediction VLA, our method surpasses CoT-VLA by 15.2%, TraceVLA by 21.5%, and DreamVLA by 3.7%. In the case of Unified VLA, our approach exceeds WorldVLA by 14.5% and UniVLA by 0.8%. This validates the effectiveness of MM-ACT in terms of model architecture and training paradigms. Furthermore, as shown in the last two rows of the table, jointly optimizing the model for both task planning and action generation during training effectively improves its capability in long-horizon planning. In particular, the success rate on Libero-Long increases from 88.0% to 93.0%, achieving a notable improvement of +5.0%.

Table 2 presents the main results of different VLA models on the RoboTwin benchmark across eight manipulation tasks. MM-ACT achieved the best average performance of

Model	Press Button	Stack Block	Sort Vegetable and Fruits	Average
π_0 [3]	75.0	70.0	65.0	70.0
OpenVLA-OFT [20]	70.0	50.0	56.0	58.6
MM-ACT	80.0	70.0	66.0	72.0

Table 3. Main results of Franka Real-world Experiments.

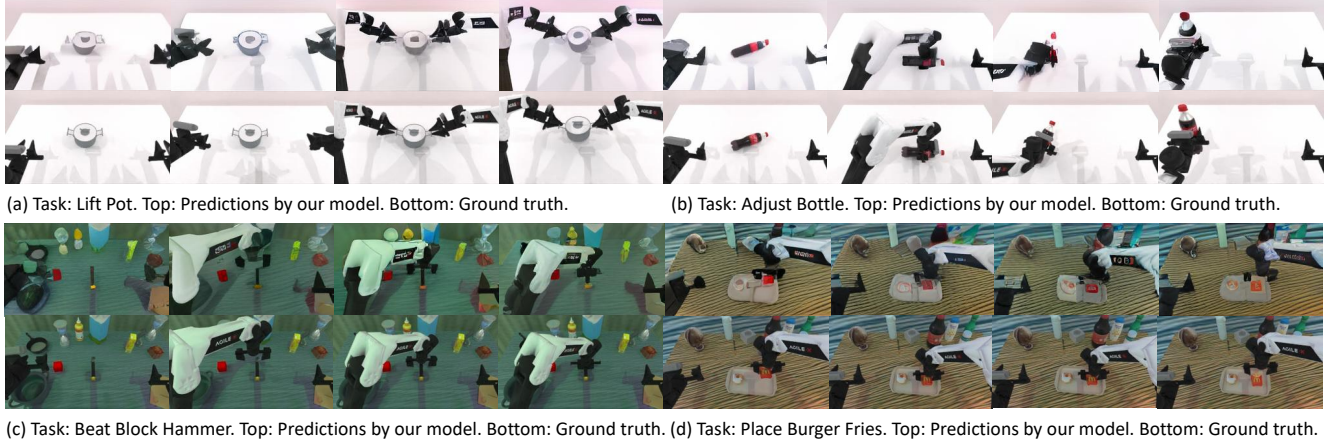


Figure 5. **Visualization of image generation by MM-ACT on RoboTwin unseen environments.** (a) and (b) show the generation results in clean scenes, (c) and (d) show the generation results in domain-randomized scenes. All of the scenes test the model’s performance in the out-of-domain setting. The top part shows the images generated by our model, while the bottom part represents the ground truth.

52.38%, surpassing π_0 by 4.25% and OpenVLA-OFT by 29.25% in terms of average success rate.

As detailed in Table 3, in the real-world Franka experiments, MM-ACT achieved the highest average success rate of 72.0%, far surpassing π_0 at 70.0% and OpenVLA-OFT at 58.6%. This clearly demonstrates the superiority of MM-ACT and further confirms its ability to successfully tackle challenging real-world tasks.

4.4. Analysis on Training Pipeline

Action Enhancement via Multimodal Learning We present the experimental results of multimodal learning in Table 1 and Table 2. Specifically, Table 1 demonstrates the effectiveness of our training pipeline, where we simultaneously train task planning and action generation on the LIBERO-Long long-horizon tasks. Compared with the baseline trained under an identical setting but generating actions only, our approach achieves a 5.0% improvement in success rate. Table 2 further compares the success rates among action-only training, text-action training, image-action training, and unified text-image-action training. Our context-shared multimodal learning approach has led to an increase in success rates for action generation, with a 3.37% improvement when co-trained with text, a 5.62% increase when co-trained with image, and a 9.25% boost when all three modalities—text, image and action—were jointly trained, achieving the highest success rate of 52.38%. This

also validates the effectiveness of our training pipeline.

Image Quality Assessment Figure 5 presents the generation results of MM-ACT (+Image) in clean and cluttered unseen scenes on RoboTwin. It can be observed that the generated images closely resemble the subgoal image, retaining key information and effectively predicting dynamic changes in the environment. More experimental results can be found in Appendix A.1.

5. Conclusion

We present MM-ACT, a unified Vision-Language-Action model that generates text, image, and robot’s action through a shared discrete token space and parallel decoding with bidirectional attention. To enhance cross-modal learning, we propose Context-Shared Multimodal Learning, which jointly supervises all modalities from the same context, fostering synergy among task planning, future image prediction and action generation. MM-ACT achieves strong results across benchmarks: 96.3% on LIBERO, 52.38% on RoboTwin2.0 eight tasks, and 72.0% on Franka real-world tasks, with context-shared multimodal learning pipeline yielding +9.25% gains in out-of-domain performance. Ablation studies validate the trade-offs between one-step parallel decoding and re-mask strategy. Overall, MM-ACT offers a compact and effective framework for discrete multimodal generation in embodied agents.

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